EE 5271 HW2: Image Registration and Tracking

This project implement feature matching, affine transform by RANSAC, image warping, and inverse compositional alignment for multi frame tracking. All core steps are coded in Python as asked in the assignment.

1) SIFT keypoints and descriptors are extracted from template and target image.
2) Matching is done by nearest neighbor ratio test, then RANSAC estimate affine matrix.
3) Target image is warped to template domain, then IC align refine the affine parameters.

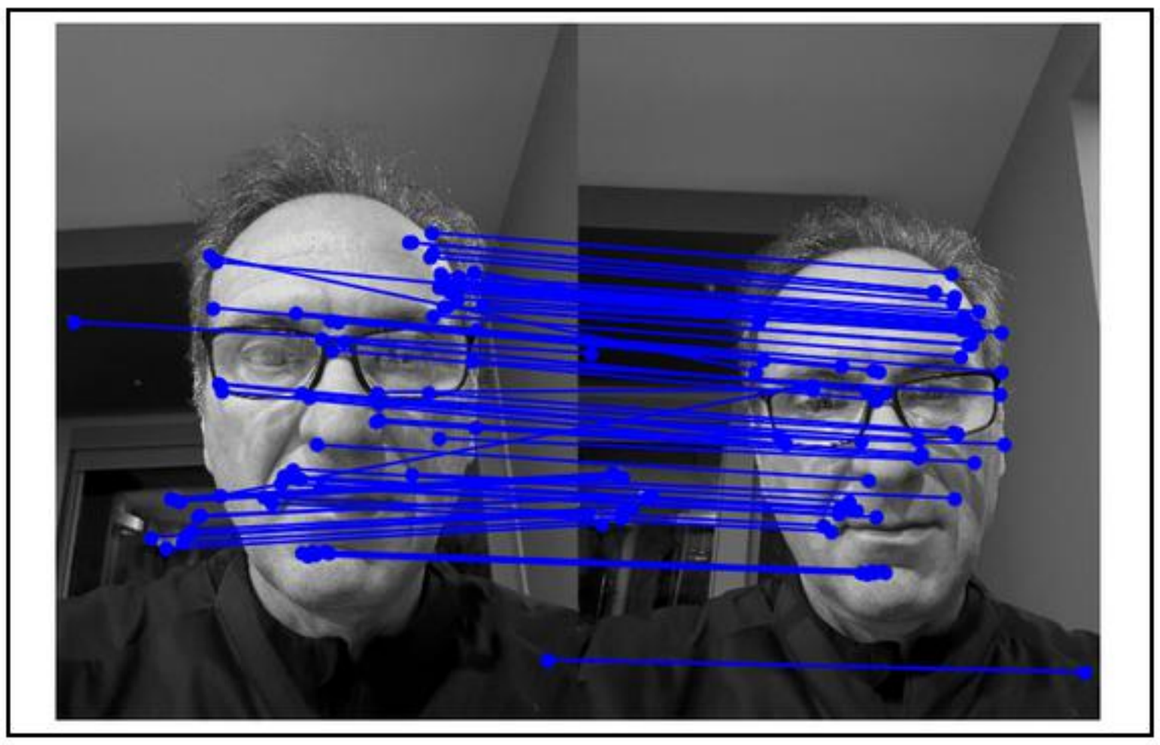

Figure 1. SIFT matches between template and frame 1.

Figure 2. Initial warped image using RANSAC affine.

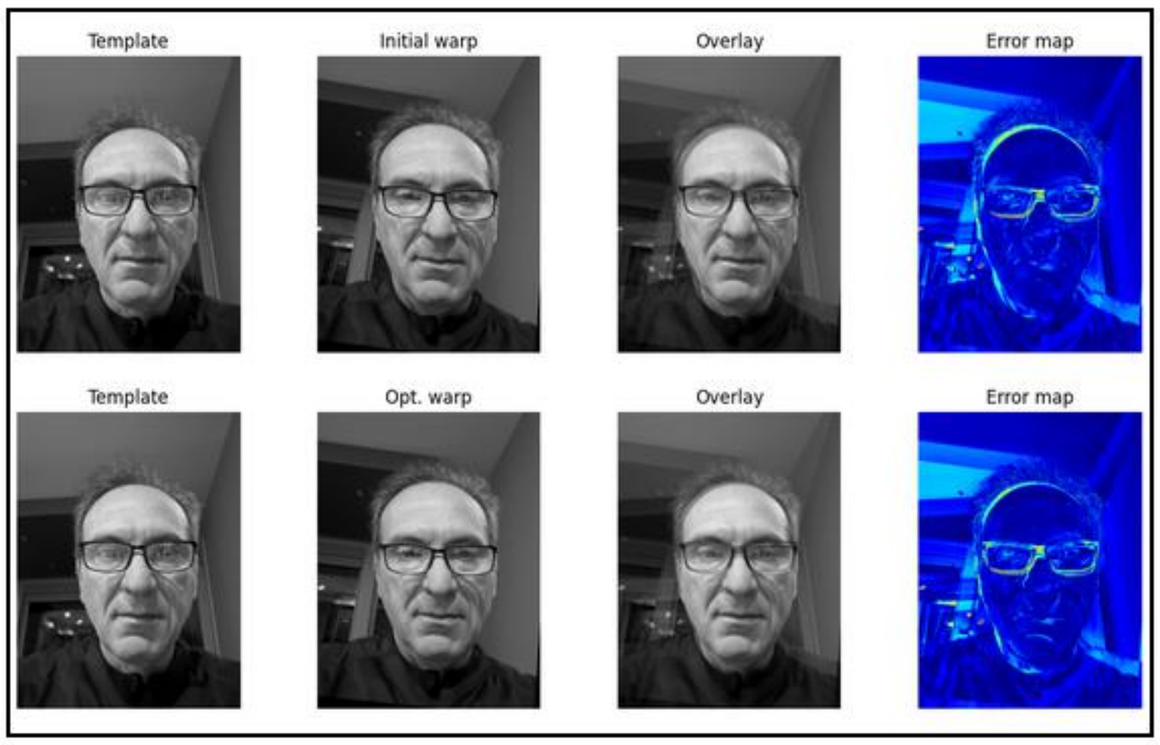

Figure 3. Before and after IC alignment.

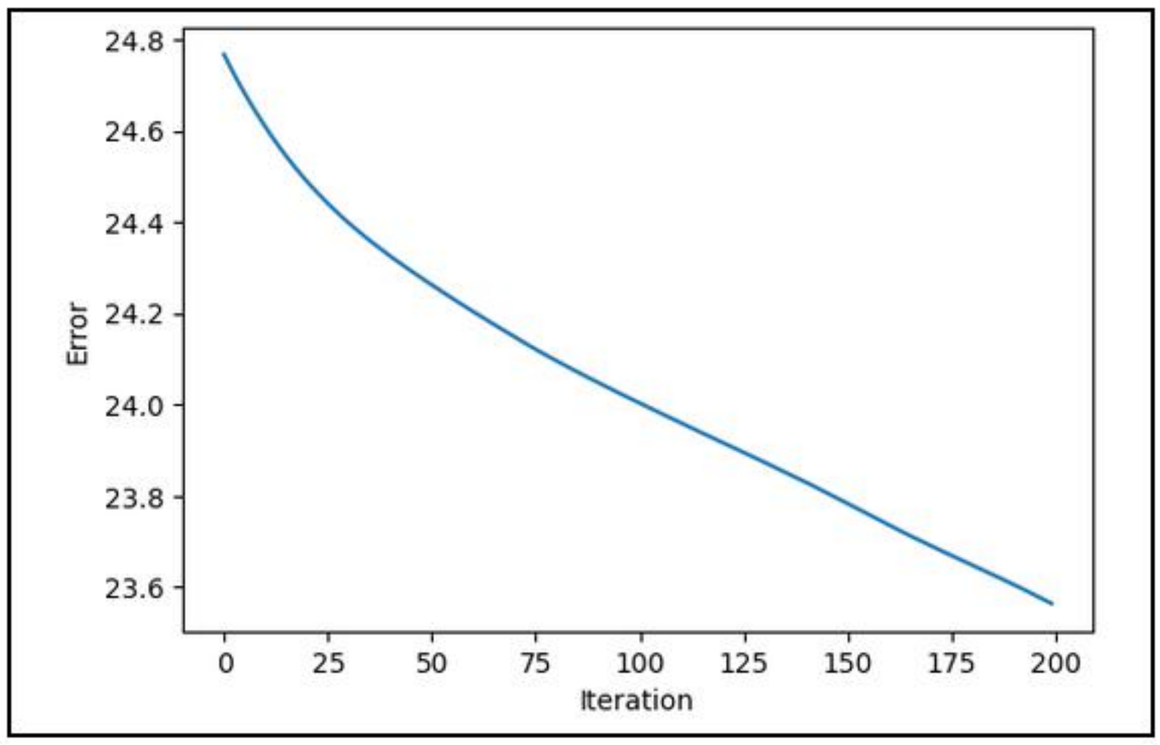

Figure 4. Alignment error go down by iterations.

Tracking Result and Discussion

For multi frame tracking, affine alignment is applied sequentially from frame 1 to frame 4. Template is updated every frame using warped current image. The red box follows head motion and rotation pretty good, but not perfect on all frames because face is 3D and affine model is 2D approximation.

The head tracking bounding box is not exactly fitting the face in all frames. Possible reasons are non-rigid face motion, outlier feature matches, and illumination or pose changes.

Figure 5. Multi frame tracking result (frames 1 to 4) with estimated affine bounding box.